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Jupyter Team_116-Q_learning_DQN_DDQN Last Checkpoint: 13 minutes ago

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JupyterLab Python [conda env:base] * 120%

All logic lives in the `src/` package so that the *same* code path is used by every experiment; this notebook only orchestrates and reports. Every module is printed in full below the section that uses it, so the exported PDF contains the complete source.

```

assignment2/
src/
  env_wrapper.py      Task (a) - StochasticActionFailureWrapper + safe-landing rule
  verify_env.py       Task (a) - V0..V3 verification suite
  agent.py            Tasks (b)/(c) - QNetwork, ReplayBuffer, DQNAgent (dqn | ddqn)
  train.py            Tasks (b)/(c) - single shared training loop + metric logging
  run_experiments.py  Tasks (b)/(c) - parallel driver over the 4 conditions x seeds
  make_plots.py       Task (d) - the four required plots + summary tables
  evaluate.py         Tasks (d)/(e) - greedy evaluation + value-estimation bias
  utils.py            seeding, moving average, fixed validation state set
  results/            per-run JSON histories, trained weights, validation states
  figures/            generated PNG figures

```

Reproduction (one command):

```

pip install "gymnasium[box2d]" torch numpy matplotlib
python -m src.verify_env                # Task (a)
python -m src.run_experiments --episodes 800 --seeds 42 43 44 # Tasks (b), (c)

```

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JupyterLab Python [conda env:base] * 120%

Reproduction (one command):

```

pip install "gymnasium[box2d]" torch numpy matplotlib
python -m src.verify_env                # Task (a)
python -m src.run_experiments --episodes 800 --seeds 42 43 44 # Tasks (b), (c)
python -m src.make_plots                # Task (d)
python -m src.evaluate                  # Tasks (d), (e)

```

[32]: %pip install numpy torch gymnasium matplotlib

```

Requirement already satisfied: numpy in /home/cloud/anaconda3/lib/python3.12/site-packages (1.26.4)
Requirement already satisfied: torch in /home/cloud/anaconda3/lib/python3.12/site-packages (2.13.0)
Requirement already satisfied: gymnasium in /home/cloud/anaconda3/lib/python3.12/site-packages (1.3.0)
Requirement already satisfied: matplotlib in /home/cloud/anaconda3/lib/python3.12/site-packages (3.10.1)
Requirement already satisfied: filelock in /home/cloud/anaconda3/lib/python3.12/site-packages (from torch) (3.13.1)
Requirement already satisfied: typing-extensions>=4.10.0 in /home/cloud/anaconda3/lib/python3.12/site-packages (from torch) (4.11.0)
Requirement already satisfied: setuptools>=77.0.3 in /home/cloud/anaconda3/lib/python3.12/site-packages (from torch) (83.0.0)
Requirement already satisfied: sympy>=1.13.3 in /home/cloud/anaconda3/lib/python3.12/site-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /home/cloud/anaconda3/lib/python3.12/site-packages (from torch) (3.3)
Requirement already satisfied: Jinja2 in /home/cloud/anaconda3/lib/python3.12/site-packages (from torch) (3.1.4)
Requirement already satisfied: fsspec>=0.8.5 in /home/cloud/anaconda3/lib/python3.12/site-packages (from torch) (2024.6.1)
Requirement already satisfied: cuda-toolkit=13.0.3 in /home/cloud/anaconda3/lib/python3.12/site-packages (from cuda-toolkit[cublas,cudart,cu
ft,cufile,cupti,curand,cusolver,cusparsa,nvjitlink,nvrtc,nvtx]=13.0.3; platform_system == "Linux"->torch) (13.0.3.0)
Requirement already satisfied: cuda-bindings<14.0.0 in /home/cloud/anaconda3/lib/python3.12/site-packages (from torch) (13.0.3.1)

```

```
[34]: # --- Notebook bootstrap: make `src` importable and record the exact runtime -----
import datetime, platform, sys, os
from pathlib import Path

# The notebook lives next to the `src` package; add its folder to sys.path.
PROJECT_ROOT = Path.cwd()
if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))

import numpy as np, torch, gymnasium as gym, matplotlib

print("run timestamp :", datetime.datetime.now().astimezone().strftime("%Y-%m-%d %H:%M:%S %Z"))
print("host / OS      :", platform.node(), "|", platform.platform())
print("python         :", sys.version.split()[0])
print("gymnasium        :", gym.__version__)
print("torch            :", torch.__version__, "| cuda:", torch.cuda.is_available())
print("numpy             :", np.__version__)
print("matplotlib        :", matplotlib.__version__)
print("project root      :", PROJECT_ROOT)
```

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JupyterLab Python [conda env:base] * ○ ≡

```
print("project root :", PROJECT_ROOT)

run timestamp : 2026-08-06 09:45:35 IST
host / OS      : 2025aa05326 | Linux-5.14.0-503.14.1.el9_5.x86_64-with-glibc2.34
python         : 3.12.7
gymnasium      : 1.3.0
torch          : 2.13.0+cu130 | cuda: False
numpy          : 1.26.4
matplotlib     : 3.10.1
project root   : /home/cloud/PycharmProjects/bits_sem2/DRL/Assignment_2/LunarLander-v3
```

[28]: def show_source(relative_path: str) -> None:
 """Print a source file verbatim so the exported PDF contains all the code."""
 path = PROJECT_ROOT / relative_path
 text = path.read_text(encoding="utf-8")
 bar = "-" * 100
 print(f"{bar}\nFILE: {relative_path} ({len(text.splitlines())} lines)\n{bar}")
 print(text)

%matplotlib inline
import matplotlib.pyplot as plt
plt.rcParams["figure.dpi"] = 110

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JupyterLab Python [conda env:base] * ⌵ ⌵ ⌵

• **Single factory.** `make_env(kind)` builds both conditions, guaranteeing that "original" and "modified" differ *only* by the presence of the wrapper.

```
[38]: show_source("src/env_wrapper.py")

=====
FILE: src/env_wrapper.py (226 lines)
=====
'''
env_wrapper.py
=====
Task (a): custom 'gymnasium.Wrapper' around **LunarLander-v3** that injects
stochastic actuator (engine) failures and a modified reward function.

Contract implemented (exactly as specified in the assignment):

    Step 1 store the action `a` selected by the agent
    Step 2 if a in {1,2,3}: draw r ~ U[0,1]; if r < 0.15 -> a_exec = 0 else a_exec = a
              if a == 0 : a_exec = 0 (unchanged)
    Step 3 execute `a_exec` in the base environment
    Step 4 R = R_base - 0.3 * 1(a != 0) + B (penalty uses the *selected* action)
    Step 5 B = 50 only for a *safe landing* (all criteria simultaneously), else 0
    Step 6 return (observation, R, terminated, truncated, info) with an **untouched** info dict
=====
```

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JupyterLab Python [conda env:base] * ⌵ ⌵ ⌵

```
heur_log, heur_eps = collect_heuristic_rollouts(n_heuristic_episodes, seed=seed + 100)

v1 = verify_misfire_rate(rnd_log, rnd_stats)
v2 = verify_fuel_penalty(rnd_log + heur_log)
v3 = verify_landing_bonus(rnd_log, rnd_eps, heur_log, heur_eps)

_rule("SUMMARY")
for name, ok in (("V0 structural equivalence", v0), ("V1 15% misfire rate", v1),
                ("V2 fuel penalty on selected action", v2), ("V3 +50 safe-landing bonus", v3)):
    print(f" {name:<38} {'PASS' if ok else 'FAIL'}")
all_ok = v0 and v1 and v2 and v3
print(f"\nOVERALL: {'ALL CHECKS PASSED' if all_ok else 'SOME CHECKS FAILED'}")
return all_ok

if __name__ == "__main__":
    raise SystemExit(0 if main() else 1)
```

```
[48]: # Run the full verification: 300 random-policy episodes + 40 heuristic-policy episodes.
from src.verify_env import main as run_verification

verification_passed = run_verification(n_random_episodes=300, n_heuristic_episodes=40, seed=7)
assert verification_passed, "environment verification failed"
```

TASK (a) - VERIFICATION OF THE MODIFIED LunarLander-v3 ENVIRONMENT

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JupyterLab Python [conda env:base] * Code

```
bootstrapped target Q-value in `_learn` - which is the only difference the
assignment allows.
"""

from __future__ import annotations

import random
from dataclasses import dataclass, asdict

import numpy as np
import torch
import torch.nn as nn
import torch.nn.functional as F

# ----- config
@dataclass
class AgentConfig:
    """All hyper-parameters in one place so DQN and DDQN provably share them."""

    hidden_sizes: tuple = (128, 128)
    lr: float = 5e-4 # Adam learning rate
    gamma: float = 0.99 # discount factor
    buffer_size: int = 100_000 # replay capacity
    batch_size: int = 64 # minibatch size
    learn_every: int = 4 # env steps between gradient updates
```